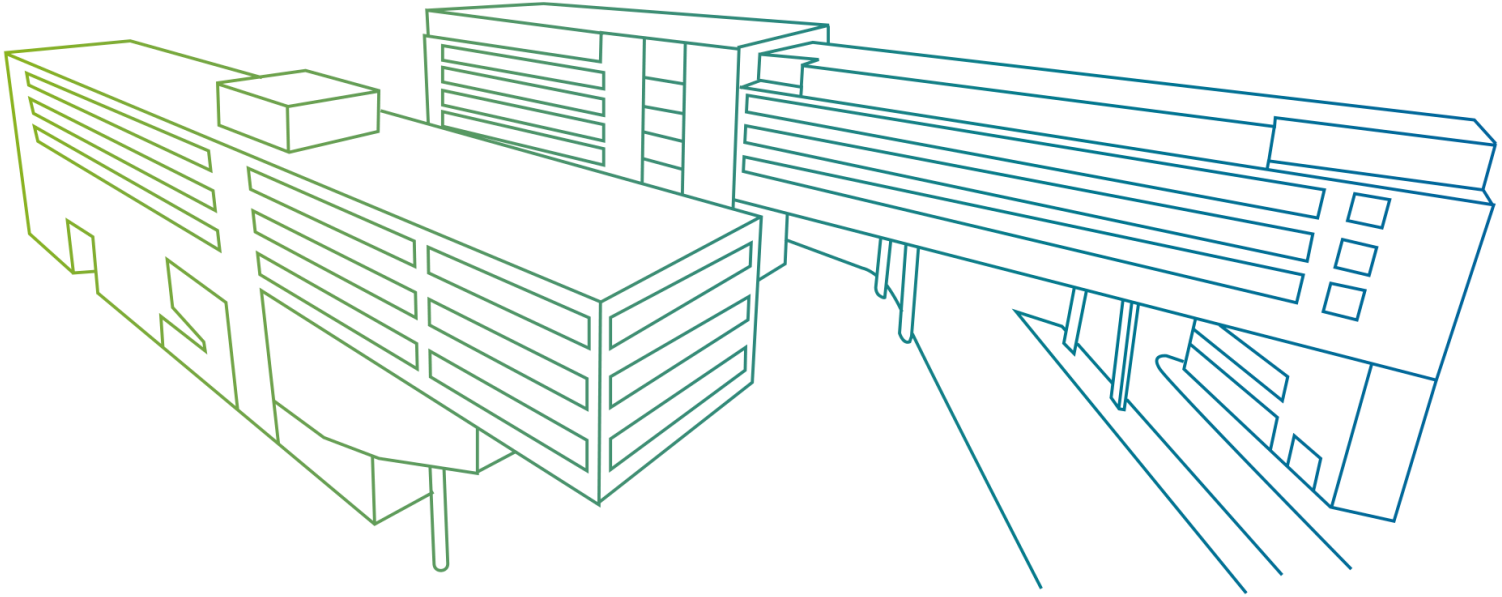


FH University of Applied Sciences TECHNIKUM WIEN

Description of Semester assignment

Alois Taranetz

Overview

- Group arrangement
- Grading
- Assignment in detail



Group arrangement

- Get together in groups in 2-4 Persons
- Every group has to announce one group leader as communication focal point for the lecturer
- How you communicate in your group is up to your personal choice
- How split up the tasks in the group is also up to your personal choice

Most important thing is that every task has a clear responsible person!

Group arrangement

- Group Leader tasks:
  - Support the Group members
  - Setup a rough time table with major milestones
  - Communicate with suppliers
  - Cost tracking
- Group Members :
  - Support the Group leader
  - Calculation of the circuit
  - Parts selection

Assignment in Detail

- Build a DC/DC non isolated converter with this spec:
- Vin: 22V-26V, Vout: 12V, Iout: 3A
- Based on LTC7800 or Silent Switcher Series
- Only Coils from Würth Eisos
- Connectors: Pin Header with 2,54mm spacing
- PCB Size: max Full Brick 116.8mm x 61mm
- Maximum 6 Layer Board allowed max. 1oz
- THT to SMD Spacing 4mm keep out
- No height restriction

Assignment in Detail

- Complies with EN 60950, UL 508, EN 61000 (EMV), CISPR16
- Output Status with LED
- Protection: Reverse Polarity at Input, Over Current at Input and Output
- Burst 2kV at Input and 1kV Surge Resistance
- Creepage, clearance and distance according to

  https://community.aisler.net/t/4-layer-1-6mm-enig-design-rules/3733
- Budget: Per Group 100 € for complete BOM
- 2-3 Working samples at the end of summer semester is the target

Group targets

- Group 1: smallest overall PCB footprint
- Group 2: cheapest switching choke
- Group 3: Highest overall efficiency
- Group 4: Best output accuracy possible
- Group 5: Thinnest overall design
- Group 6: 2-Layer PCB design
- Group 7: Only SMD parts allowed
- Group 8: Quarter Brick design

Documents to deliver at the end of the semester

- Circuit Schematic
- Time and Taskplan
- Documentation of component dimensioning
- Calculation of the part dimensions
- KiCAD exercise documents

Overall grade

Your grade will be determined based on:

- One written assignment at the end of the course with parts off all lecturer
- One presentation per group
- Quality and completeness of the created documents
- Grading description of the course also valid

Written assignment parts

- EMC lecture notes (Lamedschwandner)
- Circuit Design Script (Pruckner)
- KiCad lecture notes (Taranetz)
- learning objective control questions



- **No documents!**
- **No other aids!**
- **Don't forget your calculator!**

Good luck!

WÜRTH ELEKTRONIK
MORE THAN YOU EXPECT

